NAT and PAT Concepts Theory

Task 1: In Cisco Packet Tracer, set up a simple network with a router, one internal PC, and one external PC. Assign private IP addresses to the internal PC and public IP addresses to the external PC, then configure static NAT on the router to allow the internal PC to communicate with the external PC.

A topology was built in Packet Tracer with Router0 connecting two networks: an internal (inside) network using private addressing, and an external (outside) network simulating the public internet using addresses in the 203.0.113.0/24 documentation range.

IP Addressing

- Internal PC: 192.168.10.2 /24, gateway 192.168.10.1 (Router0 Gi0/0)
- External PC: 203.0.113.10 /24, gateway 203.0.113.1 (Router0 Gi0/1)

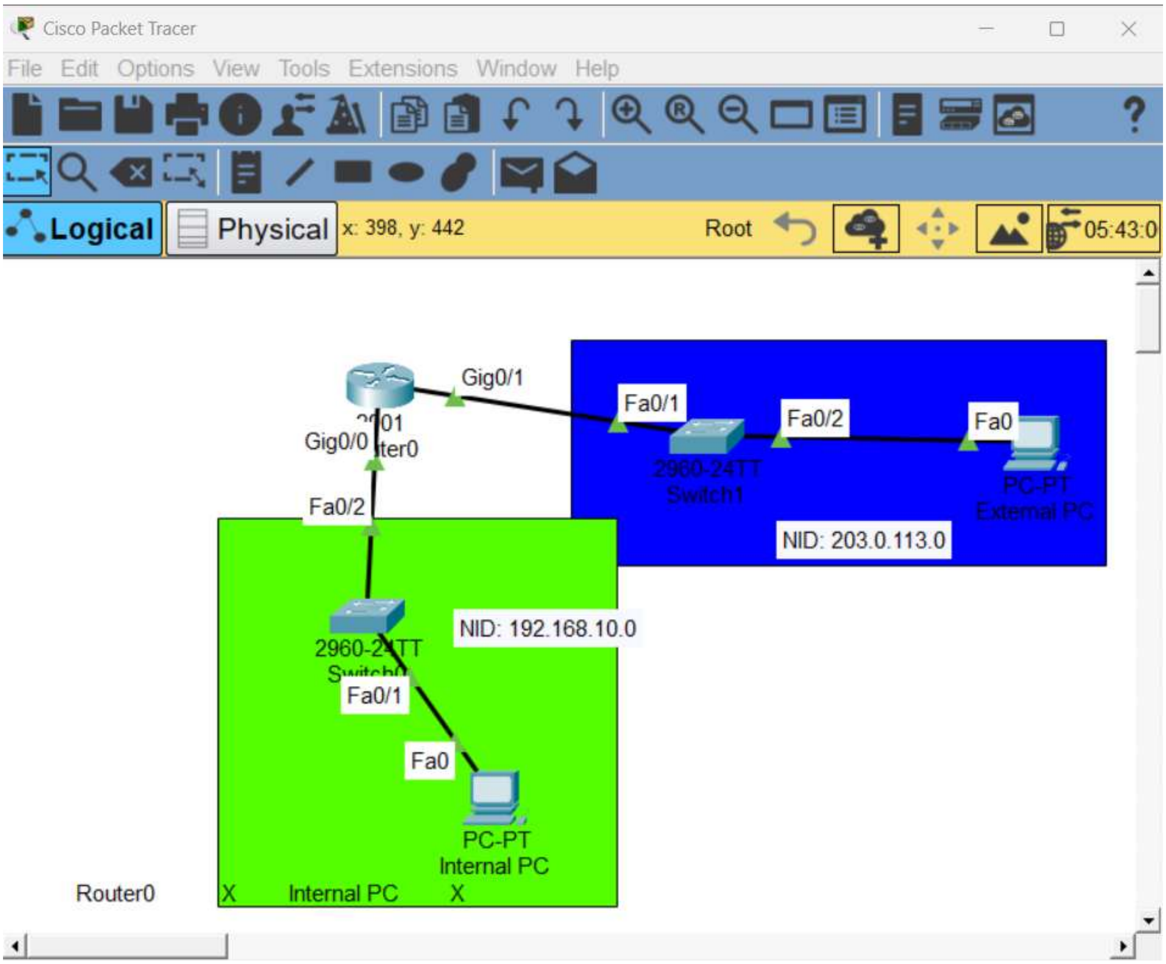


Figure 1.1 — Packet Tracer topology: internal network (green), external network (blue), Router0 in between.

Static NAT Configuration

```
Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
```

```
Router(config-if)# ip nat inside
Router(config-if)# no shutdown

Router(config)# interface GigabitEthernet0/1
Router(config-if)# ip address 203.0.113.1 255.255.255.0
Router(config-if)# ip nat outside
Router(config-if)# no shutdown

Router(config)# ip nat inside source static 192.168.10.2 203.0.113.100
```

This maps the internal PC's private address one-to-one with the public-facing address 203.0.113.100, allowing the external PC to reach it as if it had a public IP of its own.

Task 2: Configure Port Address Translation (PAT) on a router in Packet Tracer so that two internal PCs with private IPs can access a simulated public web server using a single public IP address. Hint: Use the 'ip nat inside source list' command with 'overload' to enable PAT.

The topology was extended with a second internal PC and a simulated public web server (Server0) on the external network, so that PAT could be demonstrated with multiple internal hosts sharing one translated address.

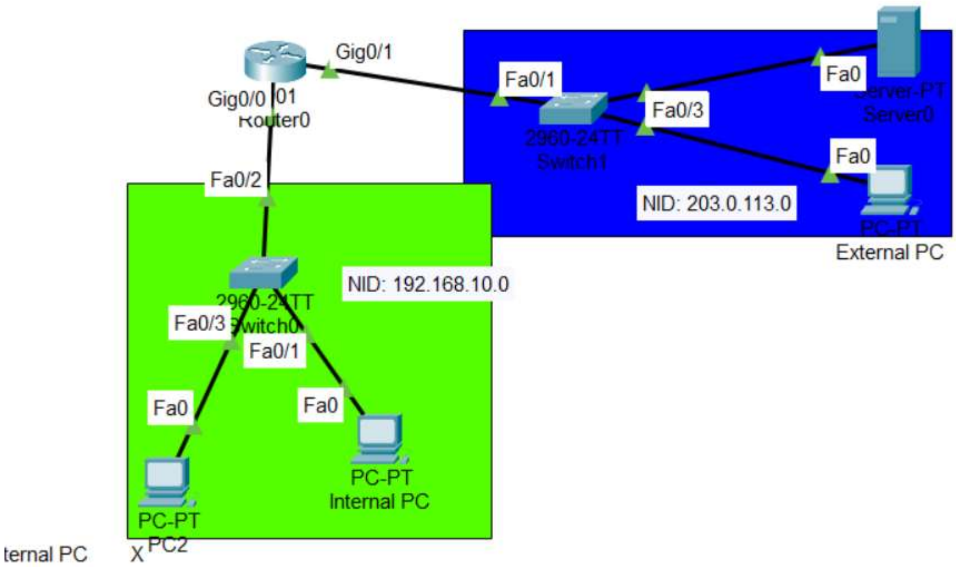


Figure 2.1 — Updated topology with a second internal PC and Server0 added to the external network.

PAT (Overload) Configuration

An access list was used to define which internal hosts are eligible for translation, and a NAT pool was created and linked with the overload keyword to enable port-based address sharing:

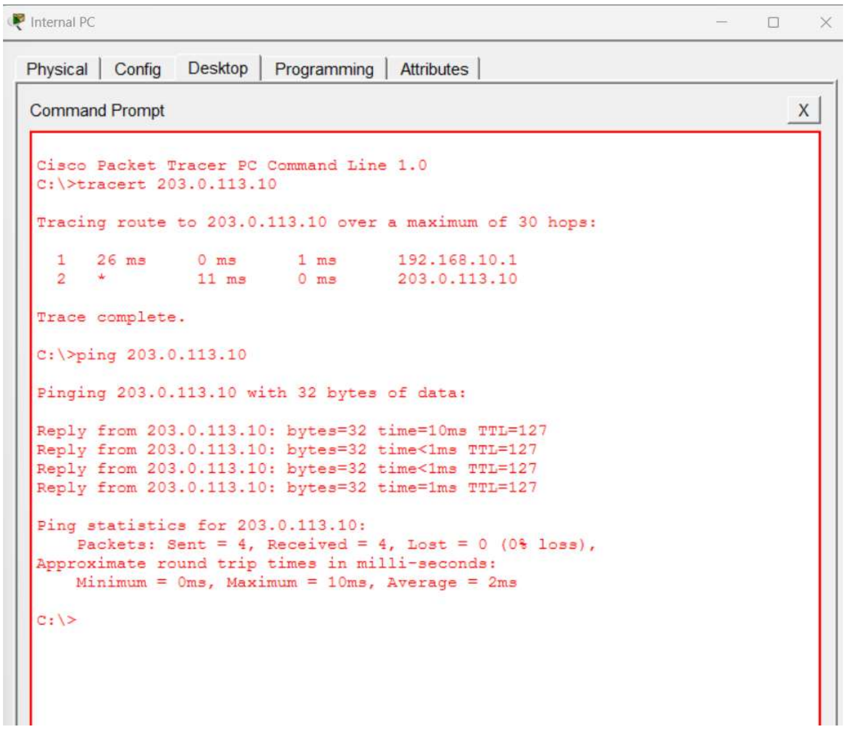
```
Router(config)# access-list 1 permit 192.168.10.0 0.0.0.255

Router(config)# ip nat pool PAT-POOL 203.0.113.200 203.0.113.203 netmask 255.255.255.0
Router(config)# ip nat inside source list 1 pool PAT-POOL overload
```

```
!
interface GigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 203.0.113.1 255.255.255.0
 ip nat outside
 duplex auto
 speed auto
!
interface Serial0/0/0
 no ip address
 clock rate 2000000
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
!
interface Vlan1
 no ip address
 shutdown
!
ip nat pool PAT-POOL 203.0.113.200 203.0.113.203 netmask 255.255.255.0
ip nat inside source list 1 pool PAT-POOL overload
ip classless
```

Figure 2.2 — Running configuration confirming NAT inside/outside interface roles, the PAT pool, and the access list.

Task 3: Test your NAT configuration by sending a ping from the internal PC to the external PC and capture the NAT translation table on the router to verify the address mapping.

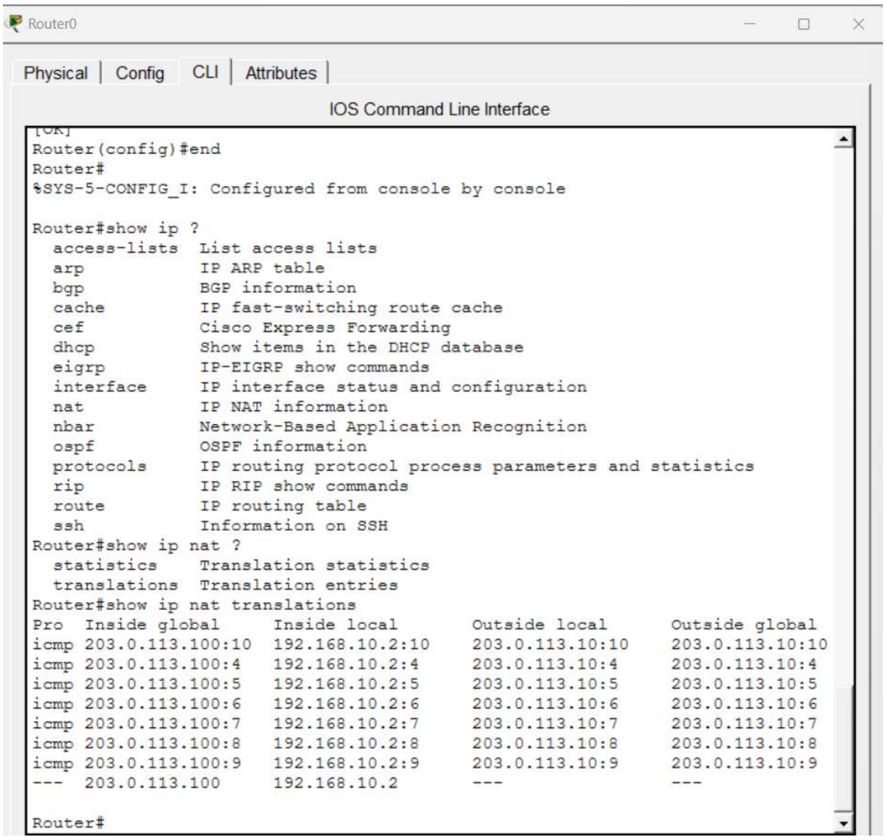
Connectivity was tested from the internal PC to the external PC using both tracert and ping, confirming end-to-end reachability through the static NAT translation configured in Task 1.



```
Internal PC
Physical | Config | Desktop | Programming | Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 203.0.113.10

Tracing route to 203.0.113.10 over a maximum of 30 hops:

  1   26 ms     0 ms      1 ms      192.168.10.1
  2   *         11 ms     0 ms      203.0.113.10

Trace complete.

C:\>ping 203.0.113.10

Pinging 203.0.113.10 with 32 bytes of data:

Reply from 203.0.113.10: bytes=32 time=10ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time=1ms TTL=127

Ping statistics for 203.0.113.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

Figure 3.1 — Successful tracert and ping (4/4 replies, 0% loss) from the internal PC to the external PC.

The router's NAT translation table was then captured to verify the address mapping in use:



```
Router0
Physical | Config | CLI | Attributes
IOS Command Line Interface
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ?
  access-lists  List access lists
  arp           IP ARP table
  bgp           BGP information
  cache         IP fast-switching route cache
  cef           Cisco Express Forwarding
  dhcp          Show items in the DHCP database
  eigrp         IP-EIGRP show commands
  interface     IP interface status and configuration
  nat           IP NAT information
  nbar          Network-Based Application Recognition
  ospf          OSPF information
  protocols     IP routing protocol process parameters and statistics
  rip           IP RIP show commands
  route         IP routing table
  ssh           Information on SSH
Router#show ip nat ?
  statistics    Translation statistics
  translations  Translation entries
Router#show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside global
icmp 203.0.113.100:10   192.168.10.2:10    203.0.113.10:10    203.0.113.10:10
icmp 203.0.113.100:4    192.168.10.2:4     203.0.113.10:4     203.0.113.10:4
icmp 203.0.113.100:5    192.168.10.2:5     203.0.113.10:5     203.0.113.10:5
icmp 203.0.113.100:6    192.168.10.2:6     203.0.113.10:6     203.0.113.10:6
icmp 203.0.113.100:7    192.168.10.2:7     203.0.113.10:7     203.0.113.10:7
icmp 203.0.113.100:8    192.168.10.2:8     203.0.113.10:8     203.0.113.10:8
icmp 203.0.113.100:9    192.168.10.2:9     203.0.113.10:9     203.0.113.10:9
---  203.0.113.100      192.168.10.2       ---                ---

Router#
```

Figure 3.2 — show ip nat translations output, confirming 192.168.10.2 translating to 203.0.113.100.

The table's final line (203.0.113.100 ↔ 192.168.10.2, with no port numbers) is the static NAT entry configured in Task 1.

Task 4: Modify your Packet Tracer topology to add a third internal PC. Update your NAT or PAT configuration so all three internal PCs can access the external network, and verify that each translation appears in the router's NAT table.

A third internal PC (192.168.10.4 /24, gateway 192.168.10.1) was added to the internal network. NAT protocol was removed, and a new access list used in the PAT rule (access-list 1 permit 192.168.10.0 0.0.0.255) already covers the entire internal subnet — any new host in that range is automatically eligible for translation.

All three internal PCs were pinged to the external web server at the same time, and the NAT translation table was captured while all three sessions were active:

```
Router#show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside global
icmp 203.0.113.200:1024192.168.10.2:5      203.0.113.20:5     203.0.113.20:1024
icmp 203.0.113.200:1025192.168.10.2:6      203.0.113.20:6     203.0.113.20:1025
icmp 203.0.113.200:1026192.168.10.2:7      203.0.113.20:7     203.0.113.20:1026
icmp 203.0.113.200:1027192.168.10.4:13     203.0.113.20:13    203.0.113.20:1027
icmp 203.0.113.200:1028192.168.10.2:8      203.0.113.20:8     203.0.113.20:1028
icmp 203.0.113.200:1029192.168.10.4:14     203.0.113.20:14    203.0.113.20:1029
icmp 203.0.113.200:1030192.168.10.2:9      203.0.113.20:9     203.0.113.20:1030
icmp 203.0.113.200:1031192.168.10.4:15     203.0.113.20:15    203.0.113.20:1031
icmp 203.0.113.200:1032192.168.10.2:10     203.0.113.20:10    203.0.113.20:1032
icmp 203.0.113.200:1033192.168.10.4:16     203.0.113.20:16    203.0.113.20:1033
icmp 203.0.113.200:1034192.168.10.2:11     203.0.113.20:11    203.0.113.20:1034
icmp 203.0.113.200:1035192.168.10.4:17     203.0.113.20:17    203.0.113.20:1035
icmp 203.0.113.200:1036192.168.10.2:12     203.0.113.20:12    203.0.113.20:1036
icmp 203.0.113.200:1037192.168.10.4:18     203.0.113.20:18    203.0.113.20:1037
icmp 203.0.113.200:1038192.168.10.2:13     203.0.113.20:13    203.0.113.20:1038
icmp 203.0.113.200:1039192.168.10.4:19     203.0.113.20:19    203.0.113.20:1039
icmp 203.0.113.200:1040192.168.10.2:14     203.0.113.20:14    203.0.113.20:1040
icmp 203.0.113.200:1041192.168.10.4:20     203.0.113.20:20    203.0.113.20:1041
icmp 203.0.113.200:1042192.168.10.2:15     203.0.113.20:15    203.0.113.20:1042
icmp 203.0.113.200:1043192.168.10.4:21     203.0.113.20:21    203.0.113.20:1043
icmp 203.0.113.200:1044192.168.10.2:16     203.0.113.20:16    203.0.113.20:1044
icmp 203.0.113.200:1045192.168.10.4:22     203.0.113.20:22    203.0.113.20:1045
icmp 203.0.113.200:1046192.168.10.2:17     203.0.113.20:17    203.0.113.20:1046
icmp 203.0.113.200:1047192.168.10.4:23     203.0.113.20:23    203.0.113.20:1047
icmp 203.0.113.200:1048192.168.10.2:18     203.0.113.20:18    203.0.113.20:1048
icmp 203.0.113.200:1049192.168.10.4:24     203.0.113.20:24    203.0.113.20:1049
icmp 203.0.113.200:1050192.168.10.2:19     203.0.113.20:19    203.0.113.20:1050
icmp 203.0.113.200:1051192.168.10.2:20     203.0.113.20:20    203.0.113.20:1051
icmp 203.0.113.200:10  192.168.10.4:10     203.0.113.20:10    203.0.113.20:10
icmp 203.0.113.200:11  192.168.10.4:11     203.0.113.20:11    203.0.113.20:11
icmp 203.0.113.200:12  192.168.10.4:12     203.0.113.20:12    203.0.113.20:12
icmp 203.0.113.200:13  192.168.10.3:13     203.0.113.20:13    203.0.113.20:13
icmp 203.0.113.200:14  192.168.10.3:14     203.0.113.20:14    203.0.113.20:14
```

Figure 4.1 — show ip nat translations while all three internal PCs (192.168.10.2, .3, .4) are pinging the external server simultaneously.

All three hosts appear in the table and are successfully reaching the external server, confirming PAT is working correctly for multiple simultaneous hosts. However, every single translation uses the same inside global address, 203.0.113.200 — the other pool addresses (.201–.203) are never touched, even though three different internal hosts are translating at once.

Why every host shares one address

With overload, IOS doesn't spread hosts evenly across the pool. It fills up all the available port numbers on the first pool address before ever moving on to the next one — and a single IP address has roughly 64,000 ports available, far more than three PCs pinging could ever use.

A simple way to picture it: imagine an apartment building with four floors, but the building manager keeps putting every new tenant on floor 1 and just hands them a different room number — 101, 102, 103 — instead of moving anyone to floor 2. Floors 2 to 4 stay empty, because floor 1 alone has room for thousands of tenants. The router only starts using the next pool address once the first one is completely full.

This is exactly the point of PAT: a single public IP address can serve a very large number of private devices, because port numbers — not additional IP addresses — are what keep each device's traffic separate and identifiable.

Task 5: Explain in your own words how NAT and PAT help apps like Zomato or Flipkart serve millions of users with limited public IP addresses. Give two real-world scenarios where incorrect NAT/PAT setup could break connectivity for users.

How NAT/PAT supports apps like Zomato or Flipkart

Public IPv4 addresses are a limited, finite resource, but millions of customer devices need to reach apps like Zomato and Flipkart every day. Most of those devices sit behind private IP addresses at home, in offices, or on mobile networks. PAT allows a whole household, office, or mobile carrier's network to share just one or a few public IP addresses when talking to these services, by using port numbers to tell each device's traffic apart — the same idea demonstrated in Task 4, where three internal PCs shared a single translated address using different ports.

This means Zomato or Flipkart's servers don't need — and realistically could never be given — a unique public IP for every single customer device. PAT lets an enormous number of users share a small pool of public addresses, which is exactly what makes large-scale internet services viable given how scarce public IPv4 addresses have become.

Two real-world scenarios where broken NAT/PAT breaks connectivity

- Port exhaustion behind an overloaded NAT: A large office or campus network shares one public IP through PAT. If too many simultaneous connections are open at once — heavy app usage, video calls, and downloads across hundreds of users — the router can run out of available ports to assign to new sessions. New connection attempts to apps like Zomato or Flipkart start failing or timing out, even though the internet connection itself is working fine.
- Misconfigured static NAT for a hosted service: If a company hosts a service internally (for example, a payment gateway or API server) and the static NAT mapping is wrong, deleted, or points to the wrong port, external users trying to reach that service get connection-refused or timeout errors — even though the server itself is running correctly. This is a common real-world cause of "it works internally but not for customers" issues.